

READER

Lesson 1 — The workbench as a test lab

AI agents on the workbench

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Introduction

KEY POINTS

- An AI agent can run the full test loop on its own.

1.1 The workbench as a test lab

The workbench as a test lab

AI agents that flash, measure and test on their own



In this lesson we build, step by step, a setup in which an AI agent can flash and test firmware on its own.

1.2 The AI agent can:

THE AI AGENT CAN:

- Flash firmware
- Watch the serial log
- Run tests
- Validate the results
- Fix problems in the code if needed

→ Workbench is an automated test lab



The agent uses ordinary command-line tools such as `esptool` and a serial monitor. See also the [ESP-IDF documentation](#).

1.3 Let the machine do the boring work, and keep the overview...

“

Let the machine do the boring work, and
keep the overview yourself.



Questions

1. Which task can the agent NOT perform on its own?

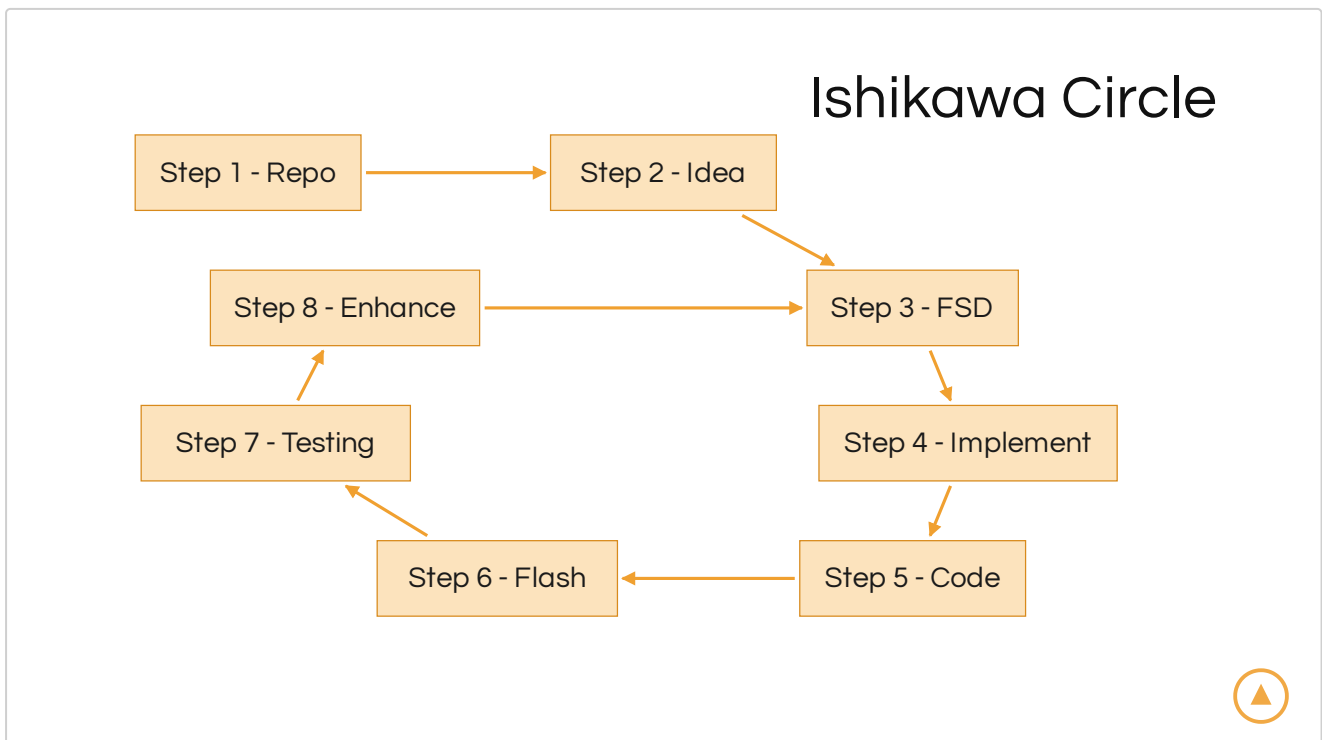
- ☐ Flash firmware
- ☐ Read the serial log
- ☐ Physically replace the board

Workflow

KEY POINTS

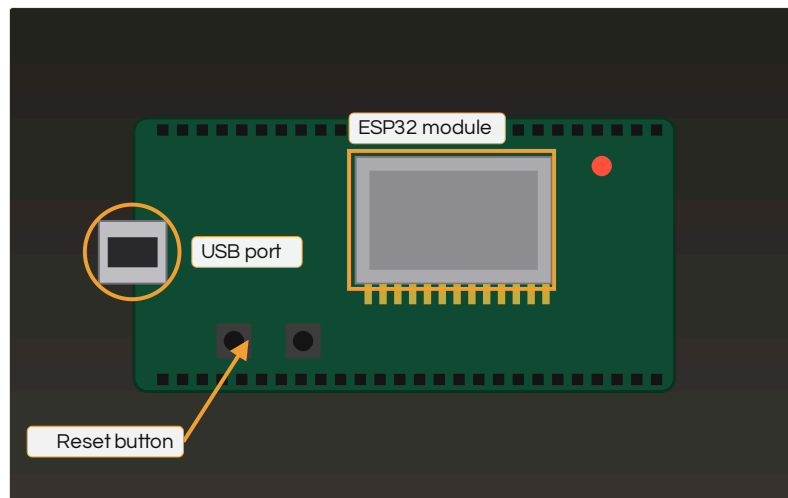
- Development is a cycle, not a straight line.
- Know the connections of your board.

2.1 Ishikawa Circle



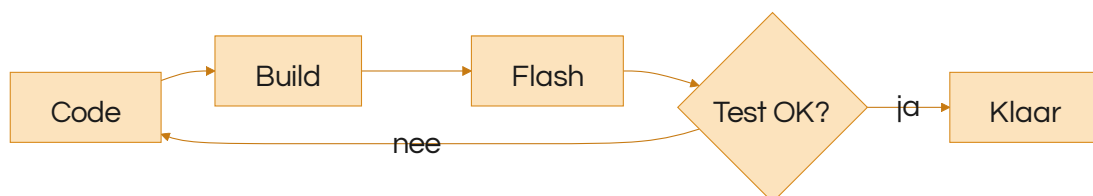
2.2 The board

THE BOARD



2.3 The test loop

THE TEST LOOP



2.4 A test in Python

A TEST IN PYTHON

```
import serial
port = serial.Serial("/dev/ttyUSB0", 115200)

def wait_for(text, timeout=10):
    line = port.read_until(text.encode(), timeout)
    return text in line.decode()

assert wait_for("BOOT OK")
```



2.5 Baud rate and time

BAUD RATE AND TIME

$$t_{byte} = \frac{10}{\text{baud rate}}$$

At 115200 baud, one byte takes about $87 \mu s$.



2.6 Manual or with an agent?

MANUAL OR WITH AN AGENT?

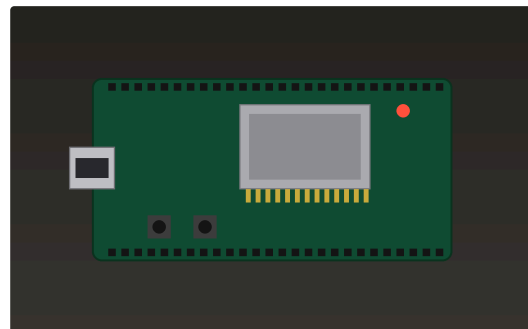
Task	Manual	With agent
Flashing	1 min	automatic
Reading logs	eyes on screen	continuous
Testing	when you remember	every build



2.7 Setup

SETUP

- Laptop with agent
- USB hub
- Test board



2.8 Thank you!

Thank you!

Questions? Comments? Let me know.

Answers

Chapter 1 — Introduction

1. Physically replace the board

Sources

1. ESP-IDF documentation
<https://docs.espressif.com/>